Day - 13
LISTS in Python

SKILL COURSE

A.I. Powered 30 Days Python Micro Course By Satish Dhawale (Microsoft Certified Trainer)

A **list** is like a bag or container where you keep **multiple items together**.

- Example:
  - Shopping list
  - To-do list
  - List of cities
  - List of marks
- **Technical Definition**
- A list is an **ordered, mutable (changeable) collection** that can store **multiple values of different data types**.
  - [item1, item2, item3]

Why Lists Are Important for Data Analysts?

Lists are the **foundation of handling tabular data** before Pandas.

Store
- Store multiple rows of data

Store
- Store names, numbers, prices, dates

Clean
- Clean entire columns using loops

Apply
- Apply transformations

Replace
- Replace wrong values

Perform
- Perform filtering

Generate
- Generate new cleaned lists

Represent
- Represent CSV columns

Lists

Creating

Creating Lists (Basic)

- fruits = ["Apple", "Banana", "Mango"]
- numbers = [10, 20, 30, 40]
- mixed = ["Satish", 25, 85.6, True]

Accessing

Accessing List Items (Indexing)

- cities = ["Mumbai", "Pune", "Delhi"]
- print(cities[0]) # Mumbai
- print(cities[1]) # Pune
- print(cities[-1]) # Delhi

Working with List

Updating List Items

- cities[1] = "Nashik"
- print(cities)

Adding Items to a List

- cities.append("Chennai")

insert() → Add at position

- cities.insert(1, "Hyderabad")

Removing Items from a List

- cities.remove("Delhi") # Remove by value
- cities.pop() # Remove last item
- cities.pop(1) # Remove by index

Working with List

Slicing

List Slicing

- nums = [10, 20, 30, 40, 50]
- print(nums[1:4]) # [20,30,40]
- print(nums[:3]) # first 3
- print(nums[-3:]) # last 3

Looping

Looping Through a List

- for item in cities:
- print(item)

Checking

Checking Membership

- print("Pune" in cities)
- print("Goa" not in cities)

Practical Example – Clean City Names

```
raw = [" mUMbai", " DELhi ", "pune ", "CHENNAI"]

clean = []

for c in raw:
    clean.append(c.strip().title())

print(clean)
```

Replacing Wrong City Spellings

```
wrong = ["Mombai", "Kolkatta", "Bengluru"]
correct = []

for city in wrong:
    city = city.replace("Mombai", "Mumbai") \
            .replace("Kolkatta", "Kolkata") \
            .replace("Bengluru", "Bengaluru")
    correct.append(city)
print(correct)
```

List of Prices → Calculate Total

```
prices = [500, 700, 1200, 850]
total = sum(prices)
print("Total Amount:", total)
```

Assignments

Basic

- Create a list of 5 fruits and print each using a loop.
- Replace any one item in the list.

Intermediate

- Clean a list of messy country names:
  [" inDIA ", " usa", "jAPAn "]
- Remove wrong names and correct spelling.

Advanced

- Extract last 4 digits from employee codes:
  ["EMP-9001", "EMP-8765", "EMP-7777"]
- Create “cleaned_products” list by removing spaces & making names title case.